

Unit-3

- prove CFLs are not closed under intersection using example.
- define DPDA, NPDA, PDA + construction & trace the string using IDs.
- ~~→ Convert DPDA to grammar~~
- Prove Pumping-Lemma for CFL.
- Convert the PDA to equivalent CFG.
- Obtain PDA with empty stack acceptance for the grammar
- ~~→ Convert the PDA to grammar~~
- define PDA & IDs
- list steps to convert CFLs to equivalent PDA by empty stack. S.T the string $a^n b^n$ is gen by the given grammar & the same is acc by eqv PDA.
- Convert CFG to eqv PDA by empty stack. S.T string $a^n a^n$ is acc by given PDA & the same is gen by equivalent grammar.
- Show that the lang is not context free.
- S.T L_1 & L_2 are context free but $L_1 - L_2$ are not context free, i.e., $L_3 = L_1 \setminus L_2$ is not context free.

→ discuss lang accepted by PDA with an ex for each. (Null, final state acceptance)

→ NPDA $\rightarrow$ CFG. (same as PDA $\rightarrow$ CFG)

→ prove CFLs are closed under Union, Concat, Closure.

→ decide if given lang is CFL or not

→ let L be $L(M_1)$ for some PDA with empty stack $M_1 = (Q, \Sigma, \Gamma, \delta, q_0, Z_0, \emptyset)$
 s.t there exists a final state PDA M_2 s.t $L = L(M_2)$.

→ appl^{ns} of CF Grammar (4 Marks)

→ let $L_1 = \{a^n b^n\}$ $L_2 = \{a^n c^n\}$ find $L_3 = L_1 \cup L_2$
 ** and $L_4 = L_1 \cap L_2$ s.t L_1, L_2, L_3 are CFLs and L_4 is not CFL.

→ Theory: ① P. Lemma for CFL.

② Appl^{ns} of CFL.

③ list steps for PDA $\rightarrow$ CFG & CFG $\rightarrow$ PDA

④ $L_1, L_2 \Rightarrow$ CFL & L_3, L_4 not C.F langs

⑤ defin PDA, NPDA, DPDA, TMs

⑥ discuss langs accepted by PDA

with an example

⑦ describe the lang acc by CFG

Unit 4 (Tmr):

- * define T.M, lang acceptance of T.M.
- * define CSL, LBA with lang acceptance
- * Construct CSG for ① $\{a^n b^n c^n \mid n \geq 1\}$
② $\{ww \mid w \in \{a, b\}^*\}$
③ $\{a^n b^n c^n d^n \mid n \geq 1\}$
ST string aabccdd is
gen by the grammar
④ $\{a^n b^n a^n \mid n \geq 0\}$.
- * Construct LBA for ① $\{a^n b^n c^n \mid n \geq 1\}$ ST
string aabba is accepted.
② $\{a^n b^n c^n \mid n \geq 0\}$
③ $\{ww^R \mid w \in \{a, b\}^*\}$
④ $N_a(w) = N_b(w) = N_c(w)$
⑤ $w \mid w \in \{a, b\}^*$
⑥ $N_a(w) = N_b(w)$

design:

- * T.M for ① $\{ww^R \mid w \text{ is any string of 0's \& 1's}\}$
② to sub 2 non-neg integers.
$$f = \begin{cases} m-n & \text{if } m \geq n \\ 0 & \text{otherwise} \end{cases}$$

③ Accept a palindrome over $\{a, b\}$
④ to perform copy operation over $\Sigma = \{a, b\}$
Trace 'bab' string.
⑤ T.M that computes the func:

$$f(x, y) = \begin{cases} x - y & \text{if } x > y \\ y - x & \text{if } y > x \\ 0 & \text{if } x = y \end{cases}$$

⑥ To perform reverse string op over $\Sigma = \{a, b, c\}$

⑦ To accept lang $\{ww \mid w \in \{a, b\}^*\}$

⑧ $\{ww^R \mid w \in \{a, b, c\}^*\}$

⑨ To compute func $f(x, y) = xy$ where $x, y \in \{a, b, c\}^*$. Trace $x = b \in ab$
 $y = abca$

⑩ quotient & remain mod 2

⑪ $N_a(w) = N_b(w)$ and $\{w \mid w \in \{a, b\}^*\}$

* Types of T.M with examples.

* P.T for a Multitape T.M there exists an equivalent single tape T.M.

* how multilimen T.M can be simulated using standard single Tape T.M.

* Define Universal T.M, NDTM

Unit - 5 (Impques):

- * ~~The PCP problem is solvable or not?~~
(Table will be given).
- * If L_1 & L_2 are RE over Σ , P.T $L_1 \cup L_2$ and $L_1 \cap L_2$ are REs.
- * define PCP, Unrestricted grammar, RE, R.L, undecidability.
- * give Ug grammar to gen:
 - ① $L = \{a^n b^n c^n \mid n \geq 1\}$. ST string aabbcc is gen.
 - ② $L = \{a^n b^n c^n d^n \mid n \geq 0\}$ & 'aabbccdd'.
 - ③ $L = \{ww \mid w \in (a,b)^*\}$.
 - ④ $L = \{w \mid w \in \{a,b,c\}^* \& N_a(w) = N_b(w) = N_c(w)\}$.
- * write a note on Chomsky hierarchy.
- * P.T. L is R.L, and so is $\bar{L}$ (complement).
- * P.T every R.L is also a R.E.L.
- * Note on undecidability.
- * P.T if L is C.S.L then L is RE.
- * P.T if L_1 & L_2 are R.L then $L_1 \cup L_2$ and $L_1 \cap L_2$ are R.Ls.